PADHAKU
Science · Class 10 Science



CHAPTER 12

Magnetic Effects of Electric Current

NCERT Exercise Solutions

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Try each question yourself first, then check the answer.
answers are samples: write your own.

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Activity 12.1

Activity 12.1 **Place a compass near a straight copper wire XY in a circuit and pass current. Observe the compass needle.**

✔ ANSWER

Observation: The compass needle is deflected when current flows through the wire.

Conclusion: An electric current through a conductor produces a magnetic field around it. Electricity and magnetism are linked.

Questions · 12.1 Magnetic Field and Field Lines

Q 1 **Why does a compass needle get deflected when brought near a bar magnet?**

✔ ANSWER

A compass needle is a small bar magnet. When it comes near a bar magnet, it is in the magnet's magnetic field.

The poles of the magnet exert forces on the poles of the needle (like poles repel, unlike poles attract), so the needle turns (deflects) and lines up along the field.

Activity 12.2

Activity 12.2 **Sprinkle iron filings around a bar magnet on a sheet of paper and tap the board gently. What do you observe?**

✔ ANSWER

Observation: The iron filings arrange themselves in a pattern of curved lines from one pole to the other; they crowd near the poles.

Conclusion: The magnet exerts a force in the region around it (magnetic field). The lines along which the filings align are the magnetic field lines.

Activity 12.3

Activity 12.3 **Draw field lines of a bar magnet using a small compass. How does the compass behave near the north pole of the magnet?**

✔ ANSWER

Observation: Near the N-pole of the magnet, the S-pole of the needle points towards it and the N-pole of the needle points away. Joining the marked points gives smooth curves from N to S of the magnet. The deflection of the needle increases near the poles.

Conclusion: These curves are the magnetic field lines. They go from the north pole to the south pole outside the magnet, and the field is strongest near the poles where the lines are crowded.

Activity 12.4

Activity 12.4 **Place a straight wire parallel to and over a compass needle and pass current, first north to south, then south to north. Observe the deflection.**

✔ ANSWER

Observation: Current north → south: the N-pole of the needle moves towards the east. Current reversed (south → north): the needle moves towards the west.

Conclusion: A current produces a magnetic field, and the direction of the field reverses when the direction of current is reversed.

Activity 12.5

Activity 12.5 **Pass current through a straight vertical copper wire through a cardboard with iron filings and tap the cardboard.**

What do the concentric circles represent? How can the direction of

the field be found? Does the direction reverse when the current is reversed?

✔ ANSWER

Observation: The iron filings form concentric circles around the wire.

- The concentric circles represent the **magnetic field lines** around the wire.
- Direction: place a compass at a point P on a circle; the N-pole of the needle shows the direction of the field there.
- **Yes**, when the current is reversed, the compass needle points the opposite way, so the field lines reverse.
- Increasing the current increases the deflection; moving the compass farther (point Q) decreases it.

Conclusion: The field around a straight current-carrying wire is concentric circles; its strength increases with current and decreases with distance from the wire.

❓ Questions · 12.2.2 Right-Hand Thumb Rule

Q 1 **Draw magnetic field lines around a bar magnet.**

✔ ANSWER

Draw a rectangle for the bar magnet with N and S at its ends.

- Draw smooth closed curves going out from the N-pole and coming into the S-pole (outside the magnet).
- Put arrows on the curves from N to S outside; inside the magnet the lines go from S to N.
- Draw the lines crowded near the poles and farther apart away from them; no two lines cross.

(See Fig. 12.4 in your book.)

Q 2 **List the properties of magnetic field lines.**

✔ ANSWER

- They emerge from the north pole and merge at the south pole (outside the magnet); inside the magnet they go from south pole to north pole.
- They are closed curves.
- They are crowded where the field is strong (near the poles) and far apart where it is weak.
- No two field lines cross each other.
- The tangent at any point gives the direction of the field (direction in which a north pole would move).

Q 3 **Why don't two magnetic field lines intersect each other?**

✔ ANSWER

If two field lines crossed, at the point of intersection the compass needle would point in two directions at the same time.

This is not possible because the magnetic field at a point has only one direction. So field lines never intersect.

Activity 12.6

Activity 12.6 **Pass current through a circular coil of many turns passing through a cardboard with iron filings; tap the cardboard and note the pattern.**

✔ ANSWER

Observation: Near each side of the wire the filings form small circles; as we go towards the centre the circles become bigger, and at the centre of the coil the lines are nearly straight and parallel.

Conclusion: All parts of a current-carrying circular coil produce field in the same direction inside the loop. The field is strong at the centre and increases with the number of turns (n turns give n times the field) and with the current.

Questions · 12.2.4 Magnetic Field due to a Current in a Solenoid

Q 1 **Consider a circular loop of wire lying in the plane of the table. Let the current pass**

through the loop clockwise. Apply the right-hand rule to find out the direction of the magnetic field inside and outside the loop.

✔ ANSWER

Hold any part of the loop with the right hand, thumb along the clockwise current. The fingers curl as follows:

- **Inside the loop:** the field is perpendicular to the table, going **into the table** (downward).
- **Outside the loop:** the field is perpendicular to the table, coming **out of the table** (upward).

Q 2 **The magnetic field in a given region is uniform. Draw a diagram to represent it.**

✔ ANSWER

A uniform field is shown by **parallel, straight, equally spaced lines** with arrows all in the same direction.

For example: draw 5–6 horizontal parallel lines at equal gaps, each with an arrow pointing to the right (like the field inside a solenoid).

Q 3 **Choose the correct option. The magnetic field inside a long straight solenoid-carrying current**

(a) is zero.

(b) decreases as we move towards its end.

(c) increases as we move towards its end.

(d) is the same at all points.

✔ ANSWER

(d) is the same at all points.

Field lines inside a solenoid are parallel straight lines, so the field is uniform.

Activity 12.7

Activity 12.7 **Suspend an aluminium rod AB between the poles of a horse-shoe magnet (field upward), pass current from B to A, then reverse it. Why does the rod get displaced?**

✔ ANSWER

Observation: With current from B to A the rod moves to the left; when the current is reversed it moves to the right. Reversing the magnetic field also reverses the movement.

Why: A current-carrying conductor placed in a magnetic field experiences a force. The direction of the force depends on the directions of current and magnetic field (Fleming's left-hand rule), and it is largest when the current is at right angles to the field.

Questions · 12.3 Force on a Current-Carrying Conductor in a Magnetic Field

Q 1 **Which of the following property of a proton can change while it moves freely in a**

magnetic field? (There may be more than one correct answer.)

(a) mass (b) speed (c) velocity (d) momentum

✔ ANSWER

(c) velocity and (d) momentum.

The magnetic force on the moving proton is perpendicular to its motion, so it changes the **direction** of motion. Velocity and momentum (which have direction) change; mass and speed do not.

Q 2 **In Activity 12.7, how do we think the displacement of rod AB will be affected if (i) current in rod AB is increased; (ii) a stronger horse-shoe magnet is used; and (iii) length of the rod AB is increased?**

✔ ANSWER

The force on the rod depends on the current, the strength of the magnetic field and the length of the rod in the field.

- (i) Current increased → force increases → **displacement increases**.
- (ii) Stronger magnet → stronger field → **displacement increases**.
- (iii) Longer rod (in the field) → force increases → **displacement increases**.

Q 3 **A positively-charged particle (alpha-particle) projected towards west is deflected towards north by a magnetic field. The direction of magnetic field is**
(a) towards south (b) towards east (c) downward (d) upward

✔ ANSWER

(d) upward.

Using Fleming's left-hand rule: current (second finger) points west (direction of the positive particle), force (thumb) points north; then the first finger (field) points **upward**.

❓ Questions · 12.4 Domestic Electric Circuits

Q 1 **Name two safety measures commonly used in electric circuits and appliances.**

✔ ANSWER

- **Electric fuse:** melts and breaks the circuit when too much current flows (short-circuit or overloading).
- **Earth wire (earthing):** connects the metal body of appliances to the earth, so leakage current goes to earth and the user does not get a severe shock.

Q 2 **An electric oven of 2 kW power rating is operated in a domestic electric circuit (220**

V) that has a current rating of 5 A. What result do you expect? Explain.

✔ ANSWER

Given: P = 2 kW = 2000 W, V = 220 V.

Current drawn: I = P/V = 2000/220 ≈ **9.1 A**.

This is much more than the 5 A rating of the circuit. So the circuit gets **overloaded**: the fuse will melt (blow) and cut off the supply; without a fuse the wires would overheat and could cause a fire.

So the oven should be used on the 15 A circuit.

Q 3 **What precaution should be taken to avoid the overloading of domestic electric circuits?**

✔ ANSWER

- Do not connect too many appliances to a single socket at the same time.
- Use high-power appliances (geyser, oven, cooler) only on the 15 A circuit.
- Use a fuse of the correct rating in each circuit.
- Keep wiring insulation in good condition to avoid short circuits.

Exercises

1 **Which of the following correctly describes the magnetic field near a long straight wire?**

(a) The field consists of straight lines perpendicular to the wire.

(b) The field consists of straight lines parallel to the wire.

(c) The field consists of radial lines originating from the wire.

(d) The field consists of concentric circles centred on the wire.

✔ ANSWER

(d) The field consists of concentric circles centred on the wire.

Iron filings around a current-carrying straight wire form concentric circles (Activity 12.5).

2 **At the time of short circuit, the current in the circuit**

(a) reduces substantially.

(b) does not change.

(c) increases heavily.

(d) vary continuously.

✔ ANSWER

(c) increases heavily.

In a short circuit the live and neutral wires touch, the resistance becomes very small, so the current rises abruptly.

3 (a) **True or false: The field at the centre of a long circular coil carrying current will be parallel straight lines.**

✔ ANSWER

True. At the centre of a current-carrying circular coil the field lines are nearly straight and parallel.

3 (b) **True or false: A wire with a green insulation is usually the live wire of an electric supply.**

✔ ANSWER

False. The green wire is the earth wire. The live wire usually has red insulation.

4 **List two methods of producing magnetic fields.**

✔ ANSWER

- Using a **permanent magnet** (e.g. a bar magnet or horse-shoe magnet).
- Passing an **electric current** through a conductor, e.g. a straight wire, a circular coil or a solenoid (an electromagnet).

5 **When is the force experienced by a current-carrying conductor placed in a magnetic field largest?**

✔ ANSWER

The force is largest when the direction of current is **at right angles (perpendicular) to the direction of the magnetic field**.

6 **Imagine that you are sitting in a chamber with your back to one wall. An electron beam, moving horizontally from back wall towards the front wall, is deflected by a strong magnetic field to your right side. What is the direction of magnetic field?**

✔ ANSWER

Electrons move from the back wall to the front wall, so the current is in the opposite direction: from the **front wall to the back wall**.

Force (deflection) is towards the right.

Fleming's left-hand rule: second finger (current) towards the back wall, thumb (force) to the right → first finger (field) points **vertically downward**.

The magnetic field is vertically downward.

7 **State the rule to determine the direction of a (i) magnetic field produced around a straight conductor-carrying current,**

(ii) force experienced by a current-carrying straight conductor placed in a magnetic field which is perpendicular to it, and
(iii) current induced in a coil due to its rotation in a magnetic field.

✔ ANSWER

(i) Right-hand thumb rule: Hold the current-carrying straight conductor in the right hand with the thumb pointing in the direction of current; the fingers wrap around the conductor in the direction of the magnetic field lines.

(ii) Fleming's left-hand rule: Stretch the thumb, forefinger and middle finger of the left hand mutually perpendicular. If the forefinger points in the direction of the magnetic field and the middle finger in the direction of current, the thumb points in the direction of force (motion) on the conductor.

(iii) Fleming's right-hand rule: Stretch the thumb, forefinger and middle finger of the right hand mutually perpendicular. If the forefinger points in the direction of the magnetic field and the thumb in the direction of motion of the conductor, the middle finger gives the direction of the induced current.

8 **When does an electric short circuit occur?**

✔ ANSWER

A short circuit occurs when the **live wire and the neutral wire come into direct contact**, e.g. when the insulation of the wires is damaged or there is a fault in the appliance.

Then the resistance becomes very small and the current increases abruptly.

9 **What is the function of an earth wire? Why is it necessary to earth metallic appliances?**

✔ ANSWER

Function: The earth wire (green insulation) is connected to a metal plate deep in the earth. It gives a low-resistance path for leakage current to flow to the earth.

Why earth metallic appliances: If current leaks to the metal body (e.g. of an electric press, toaster, refrigerator), the earth wire keeps the body at the potential of the earth and carries the leakage current away. So the user does not get a severe electric shock.

🎉 **Chapter 12 done! Revise the answers once more tomorrow.**